



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Project Overview

This comprehensive analysis examines customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

The dataset includes 18 columns covering customer demographics, purchase details, and shopping behavior metrics. Key features include age, gender, location, subscription status, item purchased, category, purchase amount, season, size, color, discount applied, promo code usage, previous purchases, frequency of purchases, review rating, and shipping type.

3,900

Purchase Records

Transactions analyzed

18

Data Columns

Features examined

Data Preparation Process

The analysis began with comprehensive data preparation and cleaning using Python and PostgreSQL.

01

Data Loading & Exploration

Imported dataset using pandas and conducted initial exploration with `df.info()` and `describe()` for summary statistics.

03

Column Standardization

Rename columns to snake case for better readability and documentation consistency.

05

Data Consistency Check

Verified `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used` column.

02

Missing Data Handling

Checked for null values and imputed 37 missing values in `Review Rating` column using median ratings by product category.

04

Feature Engineering

Created `age_group` column by binning customer ages and `purchase_frequency_days` from purchase data.

06

Database Integration

Connected Python script to PostgreSQL and loaded cleaned DataFrame for SQL analysis.



Key SQL Analysis Findings

Revenue by Gender

Male customers generated \$157,890 vs. Female customers at \$75,191

Top Products by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)

Shipping Comparison

Standard shipping: \$58.46 avg | Express shipping: \$60.48 avg

Subscription & Customer Segmentation

Subscribers vs. Non-Subscribers

Analysis revealed 1,053 subscribers generating \$62,645 in total revenue with average spend of \$59.49. Non-subscribers totaled 2,847 customers with \$170,436 in revenue and \$59.87 average spend.

Despite higher total revenue from non-subscribers, the similar average spend suggests subscription programs could be optimized to increase customer lifetime value.

Customer Segments

Customers classified into three segments based on purchase history:

- **Loyal:** 3,116 customers with consistent purchases
- **Returning:** 701 customers making repeat visits
- **New:** 83 customers with single transactions



Discount Strategy Insights

High-Spending Discount Users

839 customers used discounts but spent above average purchase amount, indicating discount strategy effectively drives higher-value transactions without sacrificing margins.

Discount-Dependent Products

Products with highest discount rates: Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%). These items rely heavily on promotions for sales.

Repeat Buyers & Subscriptions

958 repeat buyers (over 5 purchases) vs. 2,518 non-repeat buyers. Analysis suggests repeat buyers are more likely to subscribe, indicating opportunity to convert loyal customers.

Top Products by Category

The most purchased products within each category reveal customer preferences and buying patterns.



Accessories

- Jewelry: 171 orders
- Sunglasses: 161 orders
- Belt: 161 orders



Clothing

- Blouse: 171 orders
- Pants: 171 orders
- Shirt: 169 orders



Footwear

- Sandals: 160 orders
- Shoes: 150 orders
- Sneakers: 145 orders



Outerwear

- Jacket: 163 orders
- Coat: 161 orders

Revenue by Age Group

Understanding age-based spending patterns helps target marketing efforts effectively.

Young Adult

\$62,143 total revenue

Middle-aged

\$59,197 total revenue

Adult

\$55,978 total revenue

Senior

\$55,763 total revenue

Young adults generated the highest revenue, followed by middle-aged, adult, and senior segments. This insight suggests marketing campaigns should prioritize younger demographics while maintaining engagement with established customer bases.

Power BI Dashboard

An interactive dashboard was built in Power BI to present insights visually with filters for subscription status, gender, category, and shipping type.

Key KPIs

3,900 customers, \$59.76 average purchase amount, 3.75 average review rating

Visualizations

Donut chart showing 27% subscribers and 73% non-subscribers, revenue and sales by category bar charts, horizontal bar charts for age group analysis

Interactive Filters

Subscription status, gender, category, and shipping type allow dynamic exploration of data subsets

Strategic Business Recommendations

1

Boost Subscriptions

Promote exclusive benefits for subscribers to increase conversion from 27% to higher levels. Highlight perks like early access, special discounts, and free shipping.

2

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment. Create tiered programs that recognize and incentivize consistent purchasing behavior.

3

Review Discount Policy

Balance sales boosts with margin control. Consider targeted discounts on lower-margin items while maintaining full price on high-margin products like accessories.

4

Product Positioning

Highlight top-rated and best-selling products in campaigns. Feature Gloves, Sandals, and Blouses prominently based on strong performance metrics.

5

Targeted Marketing

Focus efforts on high-revenue age groups (Young Adult) and express-shipping users. Tailor messaging to demographics with highest spending patterns.